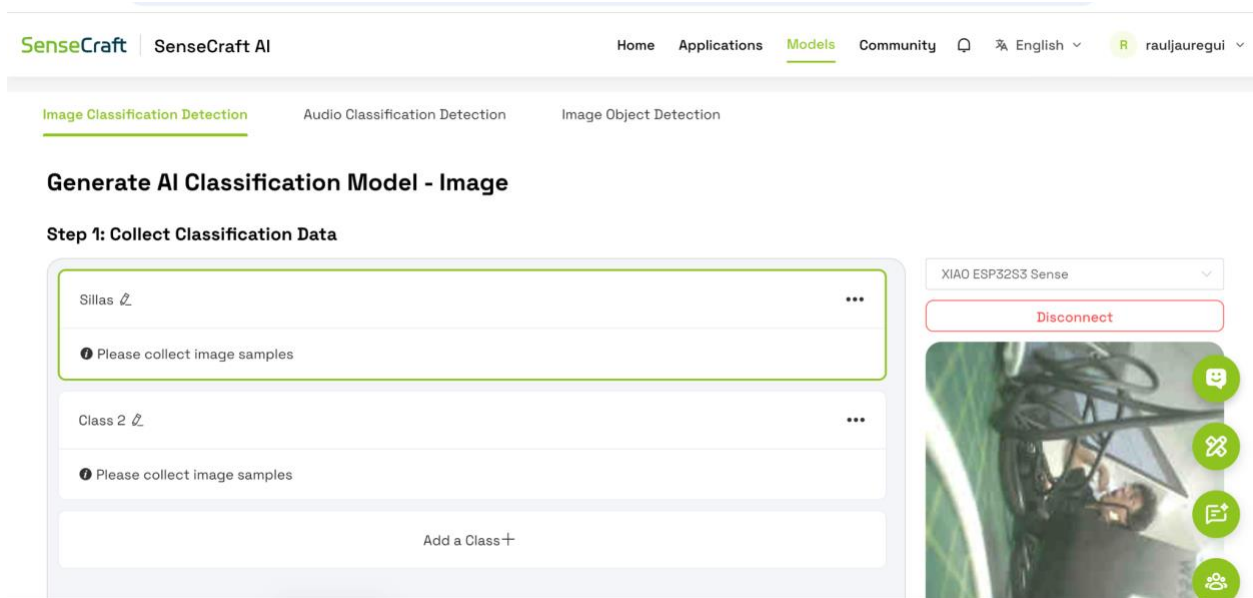


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Paso 1

Creación inicial del modelo de clasificación: Configuración inicial del proyecto en SenseCraft AI para registrar las primeras clases de objetos detectables.



Paso 2

Captura de imágenes de silla: Recolección de muestras fotográficas correspondientes a la categoría “Sillas” para entrenar el modelo.

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Step 1: Collect Classification Data

Sillas

Please collect image samples

Class 2

Please collect image samples

Add a Class

XIAO ESP32S3 Sense

Disconnect

Hold to Record
Stop

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Step 1: Collect Classification Data

Sillas

67 Image Samples

Class 2

Please collect image samples

Add a Class

XIAO ESP32S3 Sense

Disconnect

Hold to Record
Stop

Paso 3

Registro de computadoras del laboratorio: Incorporación de imágenes de computadoras y mobiliario del laboratorio para ampliar las clases del sistema.

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67 Image Samples



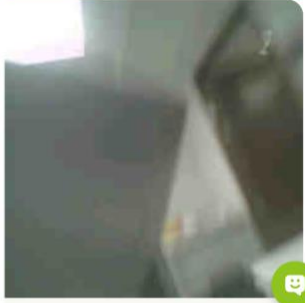
Computadoras Laboratorios 🔗

83 Image Samples



⬆️ Import Dataset⬇️ Export Dataset🗑️ Clear

Step 2: Training



Hold to RecordStop

Paso 4

Captura de imágenes de impresoras 3D: Recolección de muestras fotográficas correspondientes a la categoría “Impresoras 3D” para entrenar el modelo.

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Step 1: Collect Classification Data

Computadoras Laboratorios 🔗

125 Image Samples



Impresoras 3D 🔗

166 Image Samples



XIAO ESP32S3 Sense

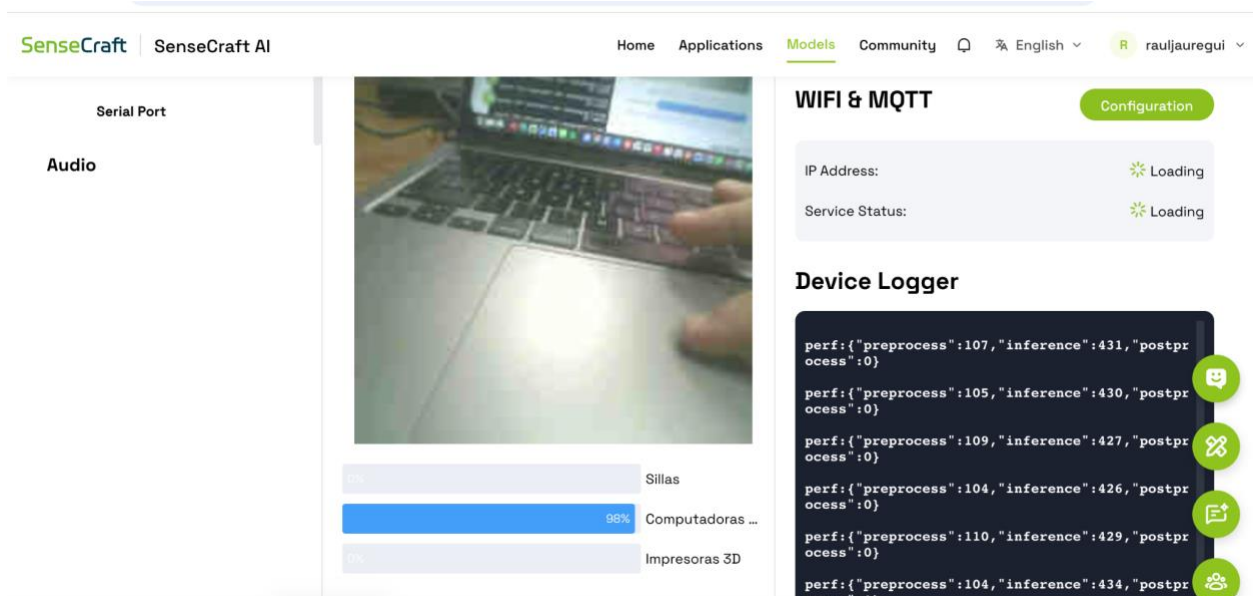
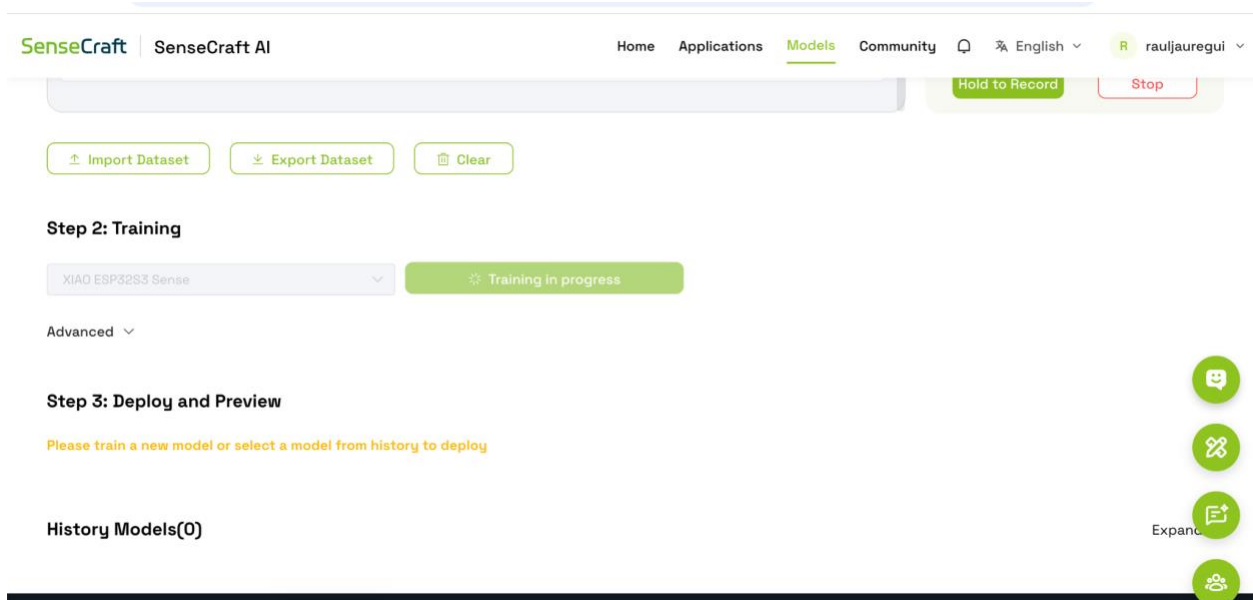
Disconnect



Hold to RecordStop

Paso 5

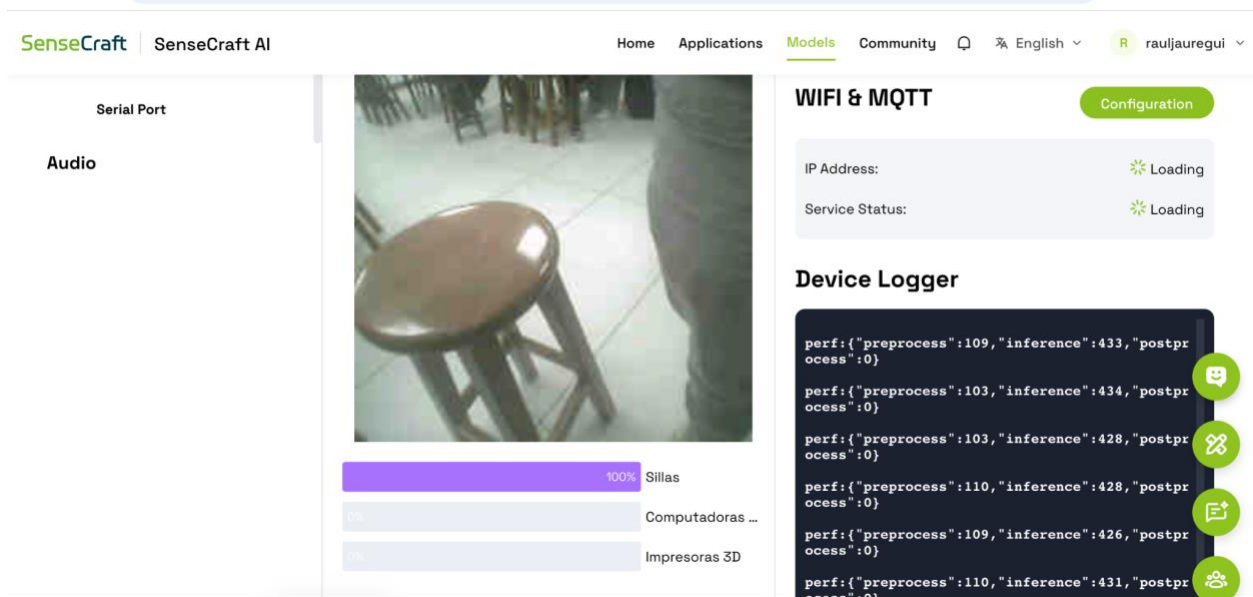
Entrenamiento preliminar del modelo: Inicio del proceso de entrenamiento utilizando las imágenes clasificadas previamente en cada categoría.



Paso 6

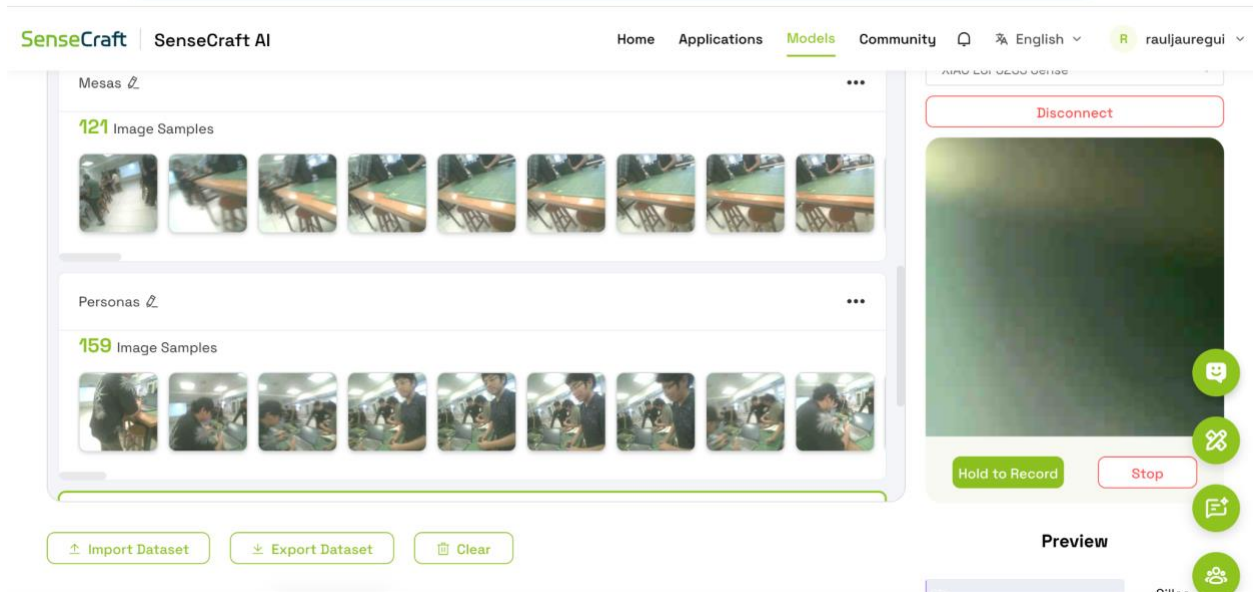
Validación de reconocimiento en tiempo real

Evaluación del modelo mediante pruebas en vivo para verificar la detección correcta de objetos.



Paso 7

Inclusión de personas y mesas: Adición de nuevas clases como personas y mesas para mejorar la capacidad de identificación del sistema.



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Audio

Sillas

Computadoras ...

100% Impresoras 3D

WIFI & MQTT
Configuration

IP Address: Loading
Service Status: Loading

Device Logger

```

perf:{"preprocess":108,"inference":432,"postprocess":0}
perf:{"preprocess":102,"inference":435,"postprocess":0}
perf:{"preprocess":100,"inference":433,"postprocess":0}
perf:{"preprocess":102,"inference":428,"postprocess":0}
perf:{"preprocess":108,"inference":428,"postprocess":0}
perf:{"preprocess":108,"inference":426,"postprocess":0}

```

Paso 8

Clasificación de objetos adicionales: Incorporación de objetos variados y pruebas de precisión para aumentar la robustez del modelo.

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Otros
184 Image Samples

Add a Class

Import Dataset
Export Dataset
Clear

Step 2: Training

Disconnect

Hold to Record
Stop

Preview

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43%

0%

48%

Impresoras 3D
Mesas
Personas

WIFI & MQTT

Configuration

IP Address: Loading
Service Status: Loading

Device Logger

```

perf:{"preprocess":77,"inference":454,"postprocess":0}
perf:{"preprocess":82,"inference":451,"postprocess":0}
perf:{"preprocess":85,"inference":452,"postprocess":0}
perf:{"preprocess":84,"inference":449,"postprocess":0}
perf:{"preprocess":83,"inference":455,"postprocess":1}
perf:{"preprocess":92,"inference":457,"postprocess":0}

```

Paso 9

Monitoreo mediante MQTT: Configuración de transmisión de datos y visualización de inferencias utilizando el protocolo MQTT.

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XIAO ESP32S3 Sense

Disconnect

Vision

Device Info

Output

MQTT

GPIO

Serial Port

Audio

Output data through MQTT

SenseCraft MQTT, which helps you quickly check whether the inference data of the device model is sent successfully

Step 1: Configure MQTT Service

Configuration

IP Address: 10.161.222.217
Service Status: MQTT connected, settings updated

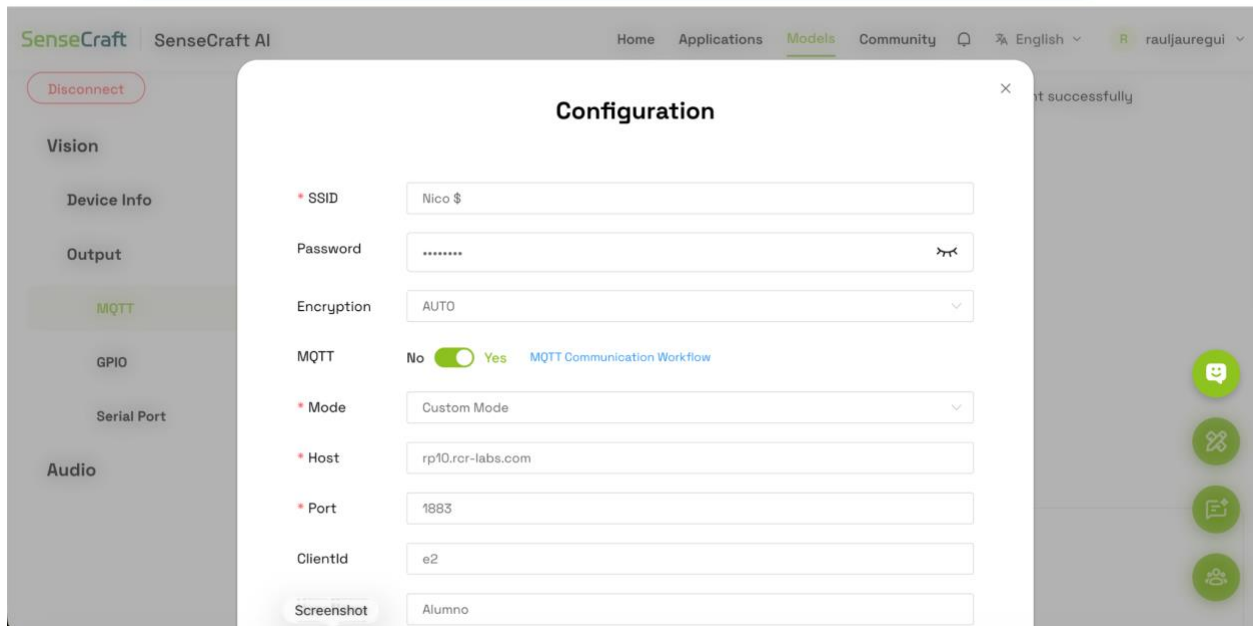
Step 2: Subscribe to the topic

Subscription: sscma/v0/Clasificacion Equipo 2/tx

Step 3: Send the wake-up command and receive messages

Paso 10

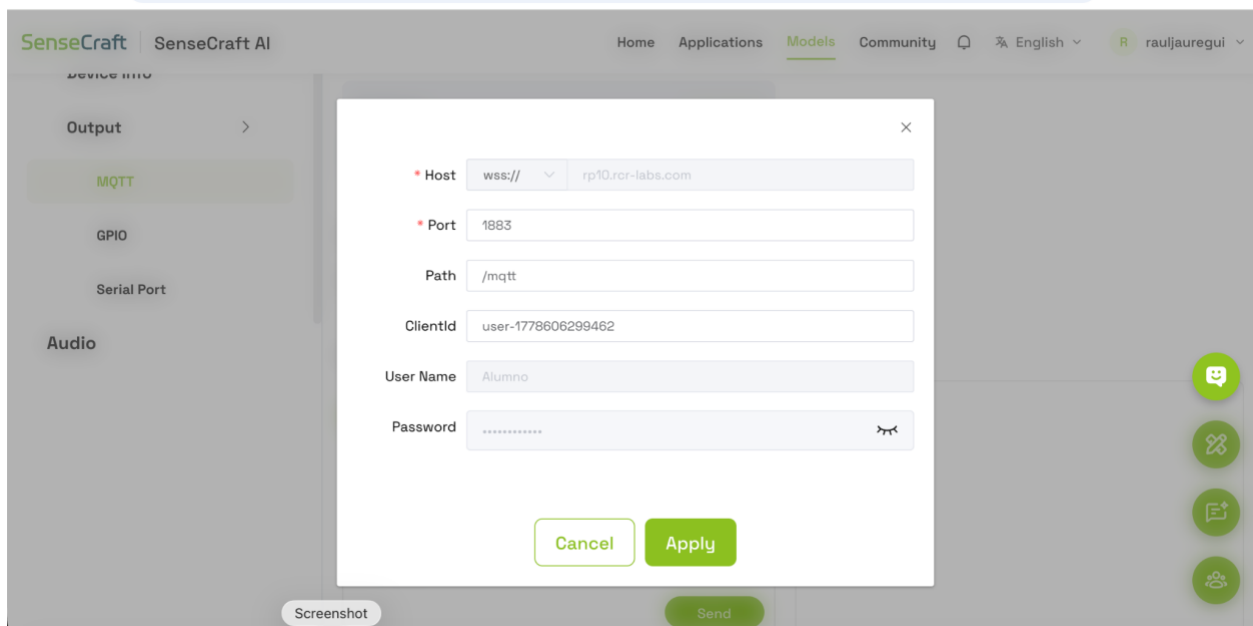
Configuración de conexión del dispositivo: Ajuste de parámetros de red, host y credenciales necesarias para la comunicación del sistema inteligente.



The screenshot shows the 'Configuration' dialog box in the SenseCraft AI interface. The dialog is titled 'Configuration' and has a close button (X) in the top right corner. It contains the following fields and options:

- * SSID:** A text input field containing 'Nico \$'.
- Password:** A text input field with masked characters (dots) and a toggle icon on the right.
- Encryption:** A dropdown menu set to 'AUTO'.
- MQTT:** A toggle switch set to 'Yes' (green), with a link 'MQTT Communication Workflow' next to it.
- * Mode:** A dropdown menu set to 'Custom Mode'.
- * Host:** A text input field containing 'rp10.rcr-labs.com'.
- * Port:** A text input field containing '1883'.
- ClientId:** A text input field containing 'e2'.
- Screenshot:** A text input field containing 'Alumno'.

The background interface shows a sidebar with 'MQTT' selected under the 'Output' section. The top navigation bar includes 'Home', 'Applications', 'Models', 'Community', and a language dropdown set to 'English'.



The screenshot shows the 'MQTT' configuration dialog box in the SenseCraft AI interface. The dialog is titled 'MQTT' and has a close button (X) in the top right corner. It contains the following fields and options:

- * Host:** A dropdown menu set to 'wss://', followed by a text input field containing 'rp10.rcr-labs.com'.
- * Port:** A text input field containing '1883'.
- Path:** A text input field containing '/mqtt'.
- ClientId:** A text input field containing 'user-1778606299462'.
- User Name:** A text input field containing 'Alumno'.
- Password:** A text input field with masked characters (dots) and a toggle icon on the right.

At the bottom of the dialog are two buttons: 'Cancel' and 'Apply'. The background interface shows the 'MQTT' section selected in the sidebar. The top navigation bar includes 'Home', 'Applications', 'Models', 'Community', and a language dropdown set to 'English'.